UNEMPLOYMENT ANALYSIS WITH PYTHON

CodeAlpha Data Science Internship – Task 2

Project Report

This report examines unemployment-rate observations in India using the two CSV datasets supplied for the CodeAlpha task. The work includes data-quality checks, preprocessing, exploratory analysis, visualisation and a descriptive comparison of observations before and during the COVID-19 period.

Item	Dataset 1	Dataset 2
Raw observations	768	267
All-empty rows	28	0
Exact duplicate rows (raw)	27	0
Clean observations	740	267
Date coverage	May 2019 – June 2020	January – October 2020
Observation level	State, month and area	State, month and region group

1. Project Objectives

• Examine unemployment-rate data and its structure.
• Identify and handle missing values, blank rows and duplicate records.
• Explore unemployment trends over time and compare rural and urban observations.
• Describe the change in unemployment during the COVID-19 period represented in the data.
• Present evidence-based findings and practical considerations.

2. Dataset Description

Two supplied datasets were retained as separate analytical sources because they have different observation structures. The first dataset contains state-level monthly observations split by Rural and Urban area, covering May 2019 to June 2020. The second dataset contains monthly state observations with a regional classification and geographic coordinates, covering January to October 2020. They were not concatenated because the row grains and schemas differ; combining them directly could double-count overlapping months or mix incompatible records.

3. Data Quality Assessment and Preprocessing

The first CSV contained 768 raw rows. It included 28 fully empty rows and 27 exact duplicate rows. After removing those records and standardising whitespace, parsing dates and validating required numeric fields, 740 records remained. The second CSV contained 267 rows and no missing values or exact duplicate rows; all 267 records were retained after standardisation and validation. The cleaned datasets have no missing values in the retained analysis columns.

Preprocessing operations included trimming whitespace from column names and text values, removing completely blank rows, removing exact duplicates, converting the Date field to datetime using day-first parsing, converting analytical fields to numeric types, normalising frequency and area labels, dropping records with invalid required

fields, and sorting the final records chronologically.

4. Exploratory Data Analysis

The second dataset's monthly average unemployment rate reached its highest level in May 2020 at 23.24% and its lowest level in October 2020 at 8.03%. The monthly series shows a sharp rise in April and May 2020, followed by a decline from June onward.

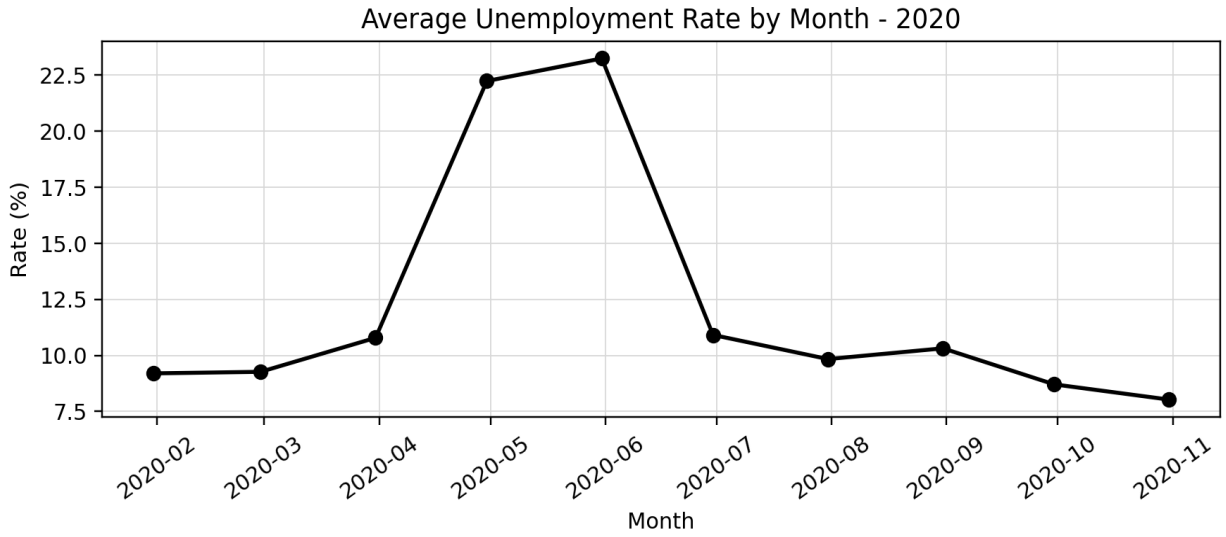


Figure 1. Monthly mean unemployment rate across the state observations in Dataset 2.

The first dataset also permits a rural–urban comparison. Both groups show a sharp increase in April and May 2020. The urban mean reached 28.40% in May 2020, while the rural mean reached 21.21% in May 2020. These are averages across the available state-area observations, not population-weighted national estimates.

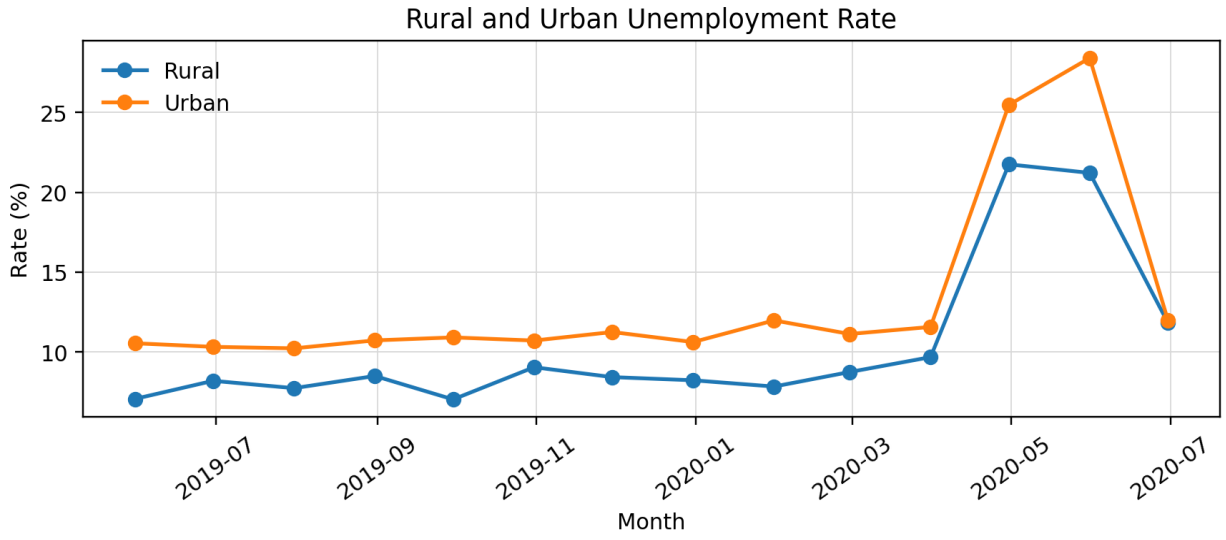


Figure 2. Monthly mean unemployment rate by area in Dataset 1.

5. COVID-19 Period Comparison

For a descriptive comparison, January and February 2020 were treated as the pre-March reference period. March to October 2020 was treated as the COVID-19 period for this analysis. The mean rate across the state-month observations was 9.23% in the reference period and 12.96% from March to October, a difference of 3.73 percentage points. The monthly mean rose to 22.24% in April and 23.24% in May, then declined to 10.91% in June.

This comparison is descriptive, not a causal estimate. The datasets do not provide a control group or enough explanatory variables to isolate the effect of COVID-19 from other economic and measurement factors. The March–October average also combines the sharp early-period increase with later months of recovery.

6. State-Level Differences

The chart below ranks the state observations by their average unemployment rate across the available January–October 2020 period. These averages are unweighted means of the available monthly observations.

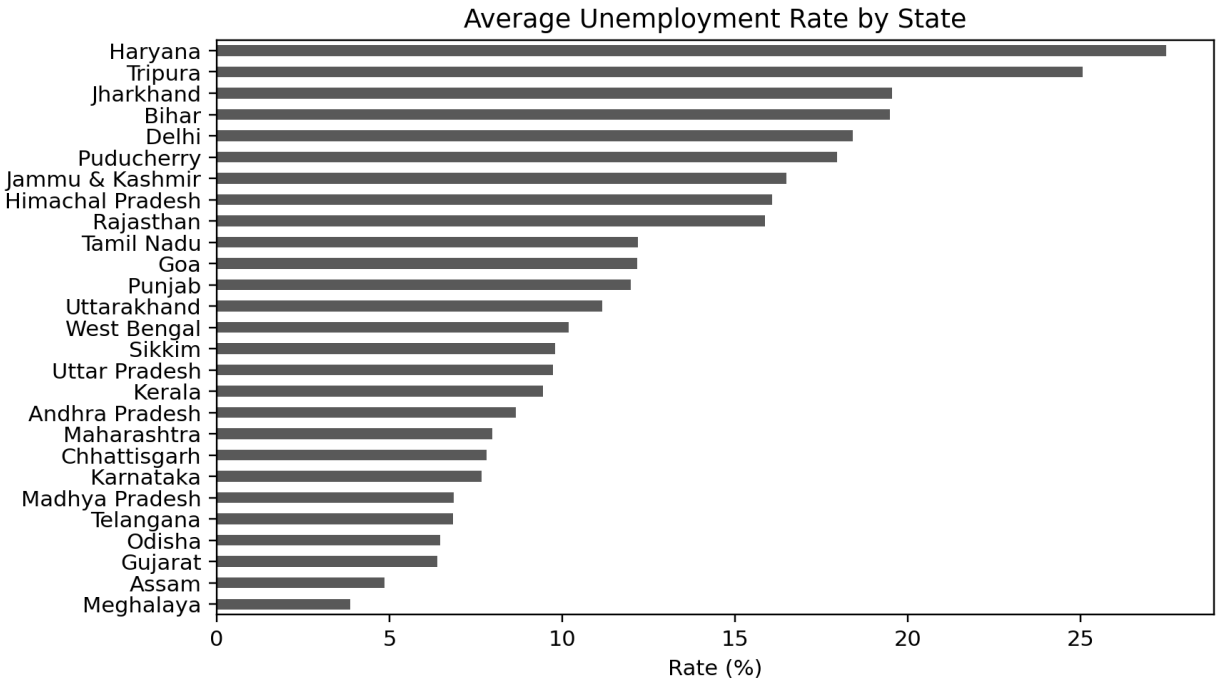


Figure 3. Average unemployment rate by state across Dataset 2.

7. Key Findings

• Dataset 1 required substantial row-level cleaning: 28 blank rows and 27 duplicate rows were removed.
• Dataset 2 had no missing or duplicate rows in the raw file.
• The state-level monthly mean peaked at 23.24% in May 2020.
• The rural–urban dataset shows the highest monthly means in May 2020 for both groups; the urban mean was higher than the rural mean in that month.
• The state-observation mean for March–October 2020 was 3.73 percentage points above the January–February reference mean.

8. Recommendations and Policy Considerations

• Monitor unemployment indicators at regular intervals and publish consistent definitions and metadata.
• Examine vulnerable regions and worker groups separately rather than relying only on a single national average.
• Pair unemployment rates with labour-force participation and employment estimates to understand whether changes reflect job loss, labour-force exit or re-entry.
• Evaluate employment-support programmes using additional data and suitable comparison methods before attributing changes to a particular intervention.

• Improve data completeness and consistency to support reliable trend and seasonal analysis.

9. Limitations

The supplied data cover a limited period and do not contain a long enough repeated annual cycle to establish robust seasonality. The datasets are observational and their estimates should not be interpreted as causal evidence. The state averages shown are not weighted by state population or labour-force size. Dataset 1 and Dataset 2 use different observation structures and have therefore been analysed separately. The analysis describes the supplied files and does not independently validate the underlying survey methodology.

10. Conclusion

The analysis identifies a pronounced rise in unemployment-rate estimates during April and May 2020, followed by a decline in subsequent months. Data cleaning improved the first dataset by removing empty and duplicate records, while the second dataset required no row removals. The results provide a clear descriptive view of unemployment patterns in the supplied data and demonstrate a reproducible Python workflow using Pandas and Matplotlib.

Tools Used

Python, Pandas, NumPy, Matplotlib and Jupyter Notebook.

Source: Two CSV datasets supplied by the intern for CodeAlpha Data Science Internship – Task 2.